

Probability Proportional to Size (PPS) Sampling

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In the case of simple random sampling, the selection probabilities are equal for all the units of the population.

If the units vary considerably in sizes, simple random sampling may not be an appropriate method because it does not give sufficient importance to large units.

To consider the importance of large units, auxiliary information can be utilized in selecting the sample to obtain more efficient estimators of population parameters.

One simple method is to assign unequal probabilities of selection to different units depending on their sizes.

PPS Sampling

- Some auxiliary characteristics x , closely related to the main characteristic y of interest, are available for all units of the population.
- For example, villages with larger geographical areas are likely to have larger populations and larger areas under food crops.
- It may be desirable to provide a sampling scheme in which villages are selected with probability proportional to their population or geographical areas.
- When units vary in size and the variate under study is highly correlated with the size of the units, PPS sampling can be used to obtain an efficient estimator.

Methods of PPS Sampling

Two different methods are used to obtain a desirable sample of a specified size:

- ① Cumulative Total Method
- ② Lahiri's Method

Cumulative Total Method

Example

Consider a village with 10 holdings consisting of the following field areas under paddy (in ropanies):

50, 30, 45, 25, 40, 26, 44, 35, 28, 27

Select a sample of four holdings with replacement using PPS sampling.

Step 1: Construct successive cumulative totals.

$$C_i = \sum_{j=1}^i X_j$$

The total size of the population is

$$X = \sum_{i=1}^{10} X_i = 350.$$

Cumulative Total Method: Calculation

Holding	Size (X_i)	Cumulative Size	Random Number Range
1	50	50	1–50
2	30	80	51–80
3	45	125	81–125
4	25	150	126–150
5	40	190	151–190
6	26	216	191–216
7	44	260	217–260
8	35	295	261–295
9	28	323	296–323
10	27	350	324–350

Selection Rule

Generate a random number R between 1 and 350. The holding whose cumulative interval contains R is selected.

Cumulative Method: Selection Procedure

Suppose the random number selected is

$$R = 272.$$

Since

$$261 \leq 272 \leq 295,$$

the random number belongs to the interval corresponding to the 8th holding.
Therefore, the 8th holding is selected.

Selection Procedure

- ① Generate a random number between 1 and 350.
- ② Identify the interval containing the random number.
- ③ Select the corresponding holding.
- ④ Repeat the process until four units are selected.

Lahiri's Method

The size of the holdings is:

50, 30, 45, 25, 40, 26, 44, 35, 28, 27.

The maximum size is

$$M = 50.$$

Procedure

- ① Select a random integer i from 1 to 10.
- ② Select another random integer j from 1 to 50.
- ③ If $j \leq X_i$, accept the i th holding.
- ④ Otherwise, reject the pair (i, j) and repeat.
- ⑤ Continue until four holdings are selected.

Lahiri's Method: Selection Table

Random i	Random j	Condition	Selected Unit
7	38	$38 < 44$	7
1	45	$45 < 50$	1
7	49	$49 > 44$	Rejected
9	38	$38 > 28$	Rejected
10	25	$25 < 27$	10
5	49	$49 > 40$	Rejected
6	32	$32 > 26$	Rejected
7	49	$49 > 44$	Rejected
4	38	$38 > 25$	Rejected
5	10	$10 < 40$	5

Selected Sample

7, 1, 10, 5

Estimation of Population Total and Variance

Consider a population of size N , and let y_i be the value of the i th unit for the study variable. Let the probability proportional to size be

$$p_i = \frac{x_i}{X}, \quad \text{where } X = \sum_{i=1}^N x_i.$$

Thus,

$$\sum_{i=1}^N p_i = 1.$$

Suppose n independent selections are made with replacement and the value of y_i for each selected unit is observed.

We consider estimation of the population mean $\bar{Y}$ and population total Y .

Hansen–Hurwitz Transformation

To estimate the population mean $\overline{Y}$, define

$$z_i = \frac{y_i}{Np_i}.$$

This is the Hansen–Hurwitz transformation.

The sample mean of the transformed variable is

$$\overline{z} = \frac{1}{n} \sum_{i=1}^n z_i = \frac{1}{n} \sum_{i=1}^n \frac{y_i}{Np_i}$$

Since z_i can take any value of the N units in the population with probability proportional to the size p_i ,

$$E(z_i) = \sum_{i=1}^N p_i z_i.$$

Unbiasedness of the Estimator

Substituting the value of z_i ,

$$E(z_i) = \sum_{i=1}^N p_i \left(\frac{y_i}{Np_i} \right) = \frac{1}{N} \sum_{i=1}^N y_i = \bar{Y}.$$

Now,

$$\begin{aligned} E(\bar{z}) &= E \left(\frac{1}{n} \sum_{i=1}^n z_i \right) = \frac{1}{n} \sum_{i=1}^n E(z_i) \\ &= \frac{1}{n} \sum_{i=1}^n \bar{Y} = \bar{Y}. \end{aligned}$$

Conclusion

The estimator $\bar{z}$ is an unbiased estimator of the population mean $\bar{Y}$.

Variance of the Estimator

The variance of z_i is

$$V(z_i) = E(z_i - \bar{Y})^2.$$

Therefore,

$$V(z_i) = \sum_{i=1}^N p_i \left(\frac{y_i}{Np_i} - \bar{Y} \right)^2.$$

Since the units are selected with replacement,

$$V(\bar{z}) = V\left(\frac{\sum_{i=1}^n z_i}{n}\right) = \frac{1}{n^2} \sum_{i=1}^n V(z_i) = \frac{\sigma_z^2}{n}.$$

Hence,

$$V(\bar{z}) = \frac{1}{n} \sum_{i=1}^N p_i \left(\frac{y_i}{Np_i} - \bar{Y} \right)^2$$

Mean and Variance

PPS Sampling

In PPS sampling with replacement, an unbiased estimator of the population mean $\bar{Y}$ is

$$\bar{z} = \frac{1}{n} \sum_{i=1}^n \frac{y_i}{Np_i}$$

and its sampling variance is

$$V(\bar{z}) = \frac{\sigma_z^2}{n} = \frac{1}{n} \sum_{i=1}^N p_i \left(\frac{y_i}{Np_i} - \bar{Y} \right)^2$$

Estimation of Population Total

An estimate of the population total Y is

$$\hat{Y}_{pps} = \frac{1}{n} \sum_{i=1}^n \frac{y_i}{p_i} = N\bar{z}$$

Taking expectation,

$$E(\hat{Y}_{pps}) = E\left(\frac{1}{n} \sum_{i=1}^n \frac{y_i}{p_i}\right) = \frac{1}{n} \sum_{i=1}^n E\left(\frac{y_i}{p_i}\right) = \frac{1}{n} \sum_{i=1}^n \sum_{i=1}^N y_i = Y.$$

Thus, $\hat{Y}_{pps}$ is an unbiased estimator of the population total.

Variance of the Estimated Population Total

$$V(\hat{Y}_{pps}) = V(N\bar{y}) = N^2 V(\bar{y})$$

$$= \frac{N^2}{n} \sum_{i=1}^N p_i \left(\frac{y_i}{Np_i} - \bar{y} \right)^2.$$

Simplifying,

$$V(\hat{Y}_{pps}) = \frac{1}{n} \sum_{i=1}^N \frac{(y_i - Yp_i)^2}{p_i}$$

Equivalently,

$$V(\hat{Y}_{pps}) = \frac{1}{n} \left[\sum_{i=1}^N \frac{y_i^2}{p_i} - Y^2 \right]$$

Thank You

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